

Global AI Startup Funding Trend Analysis (Sector/Year-wise)

Abstract

Artificial Intelligence is one of the fastest-growing areas in the technology industry. Many startups are working in different AI sectors such as Generative AI, Cybersecurity AI, Healthcare AI, Robotics, Education AI, FinTech AI and Autonomous Vehicles. These startups require funding for research, product development, hiring employees and business expansion.

This project focuses on analyzing global AI startup funding based on sector, year, country and funding stage. The project uses a dataset containing information about AI startups, their funding amount, valuation, employees, investors, country, sector and funding stage.

The data is first cleaned and analyzed using Python. Pandas and NumPy are used for data processing and calculations. MySQL and SQL are used to store the cleaned data and perform database-level analysis. Matplotlib and Seaborn are used to create graphs and visualizations. Power BI is used to create an interactive dashboard.

The main purpose of this project is to understand how AI startup funding changes over different years and sectors and how funding is distributed across different countries and investment stages.

Keywords: Artificial Intelligence, AI Startups, Startup Funding, Funding Analysis, Data Analytics, Python, MySQL, SQL, Power BI, Data Visualization

1. Introduction

Artificial Intelligence has become an important part of the technology industry. Many companies and startups are using AI to develop new products and services. AI is being used in areas such as healthcare, education, finance, cybersecurity, robotics, autonomous vehicles and Generative AI.

For startups, funding is very important because it helps them develop their products, hire employees, perform research and expand their business. The amount of funding received by startups can be different depending on their sector, country and funding stage.

Because a large amount of startup funding data is available, it can be difficult to understand the overall trend manually. Data analytics can help in organizing this information and finding useful patterns.

The project **Global AI Startup Funding Trend Analysis (Sector/Year-wise)** is developed to analyze AI startup funding using different data analytics technologies. The project mainly focuses on year-wise, sector-wise, country-wise and funding-stage-wise analysis.

2. Problem Statement

AI startup funding data contains different types of information such as startup name, country, sector, funding year, funding stage, funding amount, valuation, employees and investors.

Analyzing all these details manually is difficult and time-consuming. It is also difficult to identify which sectors receive more funding, how funding changes every year and how funding is distributed across different countries and funding stages.

Therefore, there is a need for a data analytics project that can clean the raw data, analyze it using different techniques and present the results in an easy-to-understand format.

3. Objectives

The main objectives of this project are:

1. To analyze global AI startup funding trends from 2018 to 2025.
 2. To analyze funding based on different AI sectors.
 3. To study year-wise changes in total funding.
 4. To analyze AI startup funding based on countries.
 5. To analyze funding according to different funding stages.
 6. To calculate average and total funding.
 7. To analyze startup valuation and funding amount.
 8. To create additional features such as funding per employee and funding per investor.
 9. To store and analyze the data using MySQL and SQL.
 10. To create charts and an interactive Power BI dashboard.
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4. Dataset Description

The project uses a CSV dataset containing information about global AI startups.

The raw dataset contains **1,220 records**. After performing data cleaning and removing duplicate or invalid records, the cleaned dataset contains **1,201 records**.

The dataset covers:

- 16 Countries
- 15 AI Sectors
- 7 Funding Stages
- Years from 2018 to 2025

The main columns in the dataset are:

Column	Description
Startup_ID	Unique ID of the startup
Startup_Name	Name of the startup
Funding_Year	Year in which funding was received
Country	Country of the startup
Sector	AI sector of the startup
Funding_Stage	Stage of funding
Funding_Amount_USD_Million	Funding amount in USD million
Valuation_USD_Million	Startup valuation
Employees	Number of employees
Investors_Count	Number of investors

Additional columns were created during feature engineering.

5. Technologies Used

Python

Python is used as the main programming language for the project. It is used for loading the dataset, cleaning the data, performing calculations and analyzing the results.

Pandas

Pandas is used for handling the dataset in the form of DataFrames. It is used for checking data, cleaning records, removing duplicates, grouping data and calculating summaries.

NumPy

NumPy is used for numerical calculations and handling numerical values during data processing and feature engineering.

Matplotlib

Matplotlib is used to create different charts such as year-wise funding charts and sector-wise funding charts.

Seaborn

Seaborn is used for creating statistical and analytical visualizations.

MySQL

MySQL is used to store the cleaned AI startup funding data in a structured database.

SQL

SQL is used to perform queries such as total funding, average funding, sector-wise funding, country-wise funding, year-wise funding and funding-stage analysis.

MySQL Connector/Python

MySQL Connector is used to connect the Python program with the MySQL database. Python can send SQL queries to MySQL and retrieve the required results.

Jupyter Notebook

Jupyter Notebook is used for data cleaning, exploratory data analysis, calculations and visualization.

Power BI

Power BI is used to create an interactive dashboard from the analyzed data. Users can use filters and visuals to understand funding trends.

6. Data Cleaning

Data cleaning is an important step in this project because the raw dataset may contain duplicate records, missing values, inconsistent text values and invalid numerical values.

The following operations were performed:

- Checked the number of rows and columns.
- Checked missing values.
- Checked duplicate records.
- Removed duplicate records.
- Converted numerical columns into proper numerical format.
- Checked invalid funding values.
- Handled missing categorical values.
- Handled missing numerical values.
- Standardized sector names.
- Checked country and funding-stage values.

After cleaning, the dataset contains **1,201 records**.

7. Feature Engineering

Feature engineering was performed to create some additional columns that can help in analysis.

Funding Size

The funding amount was divided into different categories to make comparison easier.

Funding Per Employee

This value shows the funding amount available per employee.

Formula:

Funding Per Employee = Funding Amount / Employees

Valuation to Funding Ratio

This value compares startup valuation with the amount of funding received.

Formula:

Valuation to Funding Ratio = Valuation / Funding Amount

Funding Per Investor

This value shows the average funding amount related to each investor.

Formula:

Funding Per Investor = Funding Amount / Investors Count

These additional features help to perform deeper analysis of startup funding.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the dataset and identify important patterns.

The analysis was performed using:

- Year
- Country
- Sector
- Funding Stage
- Funding Amount
- Valuation
- Employees
- Investors

Different group-by operations were performed to calculate total funding, average funding and startup counts.

9. Year-wise Funding Analysis

The project analyzes the total AI startup funding for each year from 2018 to 2025.

The total funding recorded for each year in the cleaned dataset is:

Year Total Funding (USD Million)

2018	2,895.57
2019	2,792.14
2020	3,414.88
2021	4,323.27
2022	3,772.38
2023	5,175.73
2024	5,112.47
2025	6,521.23

The highest total funding in the project dataset is recorded in **2025**, with approximately **USD 6,521.23 million**.

The year-wise analysis helps to understand how AI startup funding changed during the study period.

10. Sector-wise Funding Analysis

The project contains 15 different AI sectors.

Some of the sectors include:

- Cybersecurity AI
- Generative AI
- EdTech AI
- Autonomous Vehicles
- Robotics
- Healthcare AI
- FinTech AI
- Other AI-related sectors

Based on total recorded funding in the cleaned dataset, some of the major sectors are:

Sector	Total Funding (USD Million)
Cybersecurity AI	3,072.40
Generative AI	3,061.21
EdTech AI	2,995.78
Autonomous Vehicles	2,663.46
Robotics	2,500.01

This analysis helps to understand how funding is distributed across different AI application areas.

11. Country-wise Funding Analysis

The dataset contains startup funding information from 16 countries.

Country-wise analysis is performed using total funding and startup count.

Some of the countries with high recorded funding in this project dataset are:

Country	Total Funding (USD Million)
Netherlands	2,927.06
Singapore	2,851.92
Japan	2,772.78
South Korea	2,704.74
Canada	2,425.64

These values represent the records available in the project dataset and should not be treated as official national AI investment statistics.

12. Funding Stage Analysis

AI startups can receive funding at different stages depending on their business development.

The project contains the following funding stages:

- Pre-Seed
- Seed
- Series A
- Series B
- Series C
- Series D
- Late Stage

The total recorded funding by stage is:

Funding Stage	Total Funding (USD Million)
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Pre-Seed	63.62
Seed	647.96
Series A	1,909.78
Series B	3,906.50
Series C	4,760.66
Series D	8,027.10
Late Stage	14,692.05

The analysis shows that a large portion of the recorded funding in this dataset belongs to later funding stages.

13. MySQL and SQL Analysis

The cleaned data is stored in a MySQL database.

The database used in the project is:

Database: `global_ai_funding`

Table: `global_ai_startup_funding`

SQL queries are used for:

- Counting total records
- Finding available years
- Calculating total funding
- Calculating average funding
- Finding maximum funding
- Calculating total valuation
- Sector-wise funding
- Country-wise funding
- Year-wise funding
- Funding-stage analysis
- Finding top-funded startups
- Finding top countries
- Sector and year comparison

SQL makes it easier to perform structured analysis directly from the database.

14. Python–MySQL Connectivity

Python is connected to MySQL using the `mysql.connector` library.

The Python program connects to the `global_ai_funding` database and retrieves data from the `global_ai_startup_funding` table.

The database connection is tested using a Python script. This allows Python to execute SQL queries and use the results for further analysis.

This connection creates a link between the database layer and the Python analysis layer.

15. Data Visualization

Data visualization is used to present the analysis in graphical form.

The project uses Matplotlib and Seaborn to create visualizations such as:

- Year-wise funding trend
- Sector-wise total funding
- Country-wise funding
- Funding-stage distribution
- Startup count by year
- Sector-wise startup count
- Sector-year funding trends

Charts make it easier to understand large amounts of data compared with raw tables.

16. Power BI Dashboard

Power BI is used to create an interactive dashboard for the project.

The dashboard can include important KPI cards such as:

- Total Funding
- Total Startups
- Average Funding
- Total Valuation
- Total Employees
- Total Investors

The dashboard also contains charts for:

- Year-wise funding
- Sector-wise funding
- Country-wise funding
- Funding-stage analysis

- Startup count
- Sector trends

Slicers can be used for:

- Year
- Country
- Sector
- Funding Stage

This allows the user to select specific values and analyze the data interactively.

17. Results

After cleaning and analyzing the dataset, the following important results were obtained:

- Raw dataset contains **1,220 records**.
- Cleaned dataset contains **1,201 records**.
- The dataset covers **16 countries**.
- The dataset contains **15 AI sectors**.
- The dataset contains **7 funding stages**.
- The analysis covers **2018 to 2025**.
- Total recorded funding is approximately **USD 34,007.67 million**.
- Average funding per record is approximately **USD 28.32 million**.
- Maximum individual funding amount is approximately **USD 732.34 million**.
- The highest annual funding in this dataset is recorded in **2025**.
- Late Stage funding represents the largest recorded funding amount among the funding stages.

These results help in understanding the overall structure of AI startup funding in the project dataset.

18. Discussion

The analysis shows that AI startup funding changes depending on the year, sector, country and funding stage.

The year-wise analysis shows differences in total funding across the study period. Sector-wise analysis shows that different AI application areas receive different levels of funding. Country-wise analysis provides a geographical view of startup funding.

Funding-stage analysis also shows that funding amounts can become much larger in later investment stages compared with early-stage funding.

The project demonstrates that combining multiple indicators provides more information than looking only at the funding amount.

19. Limitations

The project has some limitations:

1. The dataset represents only the records included in the project.
 2. It may not include every AI startup or investment transaction worldwide.
 3. Funding values depend on the information available in the dataset.
 4. Country-wise funding values should not be considered official national investment statistics.
 5. The project mainly performs descriptive analysis.
 6. The analysis does not prove that one factor causes another.
 7. Market conditions and other external factors are not included in the dataset.
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20. Future Scope

The project can be improved in the future by:

- Adding more recent startup funding data.
 - Adding more countries.
 - Adding investor-level analysis.
 - Adding startup acquisition and IPO information.
 - Adding employee growth analysis.
 - Using machine learning for funding prediction.
 - Performing time-series forecasting.
 - Adding real-time or automated data updates.
 - Connecting Power BI directly with the MySQL database.
 - Adding more detailed AI sector classifications.
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21. Conclusion

The **Global AI Startup Funding Trend Analysis (Sector/Year-wise)** project demonstrates how data analytics can be used to understand the global AI startup funding ecosystem.

The project starts with a raw CSV dataset and performs data cleaning, feature engineering, exploratory data analysis and database analysis. Python and Pandas are used for data processing, MySQL and SQL are used for structured database analysis, and Matplotlib and Seaborn are used for visualization. Power BI is then used to present the results through an interactive dashboard.

The cleaned dataset contains 1,201 startup records covering 2018 to 2025, 16 countries, 15 sectors and 7 funding stages. The analysis provides different views of AI startup funding based on year, sector, country and funding stage.

Overall, the project demonstrates a complete practical data analytics workflow, starting from raw data and ending with meaningful analysis and interactive visualization.